WESTERN SYDNEY
UNIVERSITY

Module 8

Symbolic Math Toolbox Overview

Symbolic Math Toolbox allows you to:

- Enter expressions in symbolic form with symbolic data types.
- Expand or simplify symbolic expressions.
- Find symbolic roots, limits, minima, maxima, etc.
- Differentiate and integrate symbolic functions.
- Solve algebraic and differential equations symbolically.
- Solve simultaneous equations (even some nonlinear).
- Do variable precision arithmetic.
- Create graphical representations of symbolic functions.

Check if you have Symbolic Math Toolbox

- Type in `ver` into the Command window and it should be listed there if it is installed

- If you do not have it, follow the instructions in **this link** to install it

Symbolic Numbers

- Symbolic numbers are exact representations of numbers (unlike floating-point numbers)
- To create symbolic numbers

```
>> sym(1/3)
ans =
1/3
```

```
>> 1/3
ans =
    0.3333
```

Note: a floating-point number is decimal approximation

- A symbolic number is represented in exact rational form

Symbolic Numbers

- Calculations on Symbolic numbers are exact

```
>> sin(sym(pi))
ans =
0
```

```
>> sin(pi)
ans =
   1.2246e-16
```

- To add symbolic numbers

```
>> sym(1/3)+sym(5/7)

ans =

22/21
```

Symbolic Variables

- To create a symbolic variable use

```
syms x
```

This makes a symbolic variable `x` with value `x` assigned to the variable `x`

Creating Multiple Variables

- To create several symbolic variables in one line

```
syms K T P0;
```

This creates the symbolic variables **`K`**, **`T`**, and **`P0`**; (check using the command **whos** command)

Creating a symbolic expression with symbolic variables

- To create an expression using existing symbolic variables:

```
P = P0*exp(K*T)
```

- This creates a symbolic expression that includes the exponential function.

Symbolic equations

- It is possible to write equations with the symbolic toolbox. Example:

```
ideal_gas_law = P*V==n*R*Temp
```

Substitution with subs()

- The subs() command allows you to substitute a symbol with another symbol or assign a number to a variable.

```
syms a b c x y;
quadratic = a*x^2 + b*x + c;
yquadratic = subs(quadratic,x,y)
```

```
yquadratic =
a*y^2 + b*y + c
```

Multiple substitutions

- You can use the **`subs()`** command to do multiple substitutions in one command. This is done by grouping the variables and their substitutes (other variables or numerical values) in braces.

```
subs(symbolic_function, [variables], [substitute])

subs(quadratic, [a,b,c,x], [1,2,3,4])

ans =
27
```